

MATH 20700 Honors Analysis I

Homework 7

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Problem 5.15

Proof.

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f((0+h,0)) - f((0,0))}{h} = \lim_{h \rightarrow 0} \frac{0-0}{h} = 0.$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{h \rightarrow 0} \frac{f((0,0+h)) - f((0,0))}{h} = \lim_{h \rightarrow 0} \frac{0-0}{h} = 0.$$

The total derivative, if it exists, is given by

$$f(p+v) = f(p) + (Df)_p(v) + R(v),$$

with the error term $R(v)$ sublinear.

Let $v = (r, s)$. Let us approach $v \rightarrow (0,0)$ along the line $y = x$ i.e., $r = s$. We get $f(r, r) = \frac{r^2}{2r^2} = \frac{1}{2}$. Hence, $\lim_{v \rightarrow 0} f(v) = \frac{1}{2} \neq 0$, hence f is not continuous at the origin, hence it is not differentiable at the origin.

QED

Problem 5.17

Proof. (a) Remembering that Df is simply the Jacobian matrix of partials, we get that $Df = \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}$.

Observe that $f(q) = f(2\pi) = (1,0)$ and $f(p) = f(\pi) = (-1,0)$. Thus, $f(q) - f(p) = (2,0)$ and $q - p = \pi$.

Suppose for sake of contradiction that $\exists \theta \in [p, q]$ that satisfies equation 28. Then, we would have $(Df)_\theta \cdot \pi = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$. Thus, $(Df)_\theta = \begin{pmatrix} \frac{2}{\pi} \\ 0 \end{pmatrix}$. Thus, $-\sin \theta = \frac{2}{\pi}$ and $\cos \theta = 0$.

In the interval $[\pi, 2\pi]$, there is a unique point $t = \frac{3\pi}{2}$ such that $\cos t = 0$. And we observe that $-\sin(\frac{3\pi}{2}) = 1 \neq \frac{2}{\pi}$.

Hence we conclude that $\nexists \theta \in [p, q]$ that satisfies equation 28.

(b)

(c)

QED

Problem 5.18

Proof. (a) If f is differentiable at p , we have $f(p+v) - f(p) - (Df)_p(v) = R(v)$ with $R(v)$ sublinear, i.e., $\lim_{v \rightarrow 0} \frac{\|R(v)\|}{\|v\|} = 0$.

Let $v := tu$. Then, we have $f(p+tu) - f(p) - (Df)_p(tu) = R(tu)$ with $R(tu)$ sublinear, i.e., $\lim_{tu \rightarrow 0} \frac{\|R(tu)\|}{\|tu\|} = 0$.

Let us divide by t throughout, then we get

$$\frac{f(p+tu) - f(p) - (Df)_p(tu)}{t} = \frac{f(p+tu) - f(p) - t \cdot (Df)_p(u)}{t} = \frac{R(tu)}{t}.$$

Hence, $\frac{f(p+tu) - f(p)}{t} = (Df)_p(u) + \frac{R(tu)}{t}$.

Hence, $\lim_{t \rightarrow 0} \frac{f(p+tu) - f(p)}{t} = \lim_{t \rightarrow 0} (Df)_p(u) + \lim_{t \rightarrow 0} \frac{R(tu)}{t} = (Df)_p(u) + 0$.

This shows that the limit $\lim_{t \rightarrow 0} \frac{f(p+tu) - f(p)}{t}$ exists, hence the directional derivative $\nabla_p f(u)$ exists.

(b) Writing $u = (x, y)$, so that $tu = (tx, ty)$, we get

$$\nabla_{(0,0)} f(u) = \lim_{t \rightarrow 0} \frac{f(tu)}{t} = \lim_{t \rightarrow 0} \frac{(tx)^3(ty)}{((tx)^4 + (ty)^2)t} = \lim_{t \rightarrow 0} \frac{t^4 x^3 y}{t^5 x^4 + t^3 y^2} = \lim_{t \rightarrow 0} \frac{tx^3 y}{t^2 x^4 + y^2} = 0.$$

If f were differentiable at $(0,0)$, the derivative $(Df)_{(0,0)}$ would have to be the Jacobian matrix of partial derivatives, and this is exactly the zero matrix, since $\forall u : \nabla_{(0,0)} f(u) = 0$.

And the derivative $(Df)_{(0,0)}$ would have to satisfy $f(v) - (Df)_{(0,0)}(v) = f(v) = R(v)$ with $R(v)$ sublinear.

This is a contradiction since $f(v)$ cannot be of the same order as the error term $R(v)$ if f is differentiable.

We must now show that $\lim_{v \rightarrow 0} \frac{\|f(v)\|}{\|v\|} \neq 0$. Let $v = (x, y)$. We need to test the limit

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\left| \frac{x^3 y}{x^4 + y^2} \right|}{\sqrt{x^2 + y^2}}.$$

We try approaching $(0,0)$ along the parabolic path $y = x^2$, i.e., $v = (x, x^2)$, then $\|v\| = x\sqrt{1+x^2}$. $f(x, x^2) = \frac{x^5}{2x^4} = \frac{x}{2}$. Hence, $\lim_{v \rightarrow 0} \frac{\|f(v)\|}{\|v\|} = \lim_{v \rightarrow 0} \frac{x/2}{x\sqrt{1+x^2}} = \lim_{v \rightarrow 0} \frac{1}{2\sqrt{1+x^2}} = \frac{1}{2} \neq 0$.

QED

Problem 5.20

Proof. Since f is differentiable and $\forall p \in U : (Df)_p = 0$, we know that the total derivative (Jacobian) matrix is the zero matrix. Hence, all the directional derivatives must be zero. Every directional derivative can be written as a linear combination of the partial derivatives, hence each of the partial derivatives $\frac{\partial f_i}{\partial j}$ must be zero.

If f is not constant, then there must be at least one component function out of $\{f_1, f_2, \dots, f_m\}$ that is not constant.

Suppose f_i is not constant, so this means that $\exists w_1 \neq w_2 \in \mathbb{R}^m$ such that $w_1 = f(v_1), w_2 = f(v_2)$. Clearly $v_1 \neq v_2$, so there must be some component for which v_1 and v_2 differ, call it j (i.e., $v_{1j} \neq v_{2j}$).

Then, $\frac{\partial f_i}{\partial v_j} \neq 0$, contradicting the fact that each of the partial derivatives must be zero.

We note that the assumptions in the problem statement are important: we want the set $U \subset \mathbb{R}^n$ to ‘behave like the real numbers’ (e.g., like a Cartesian product of intervals).

Continuity is a necessary (but not sufficient) condition for differentiability. Hence we want limits to converge under continuous functions, thus we want U to be connected.

And we want the derivative to be defined in the neighbourhood of every point $p \in U$, hence we want U to be open.

QED

Problem 5.22

Proof. Fix $y_0 \in Y$ and let $\epsilon > 0$. We want to show there exists a $\delta > 0$ such that for all y with $d_Y(y, y_0) < \delta$:

$$\int_a^b |f(x, y) - f(x, y_0)| dx < \epsilon$$

It suffices to find a δ such that for all $x \in [a, b]$:

$$|f(x, y) - f(x, y_0)| < \frac{\epsilon}{b - a}$$

Since f is continuous, it is continuous at every point (x, y_0) where $x \in [a, b]$. For any specific $x \in [a, b]$, continuity at (x, y_0) implies there exist open neighborhoods:

1. $U_x \subset [a, b]$ containing x (an open interval), and
2. $V_x \subset Y$ containing y_0 (an open ball of radius δ_x),

such that for any $(s, t) \in U_x \times V_x$:

$$|f(s, t) - f(x, y_0)| < \frac{\epsilon}{2(b - a)}$$

If we take any $s \in U_x$ and any $y \in V_x$, we can use the Triangle Inequality:

$$|f(s, y) - f(s, y_0)| \leq |f(s, y) - f(x, y_0)| + |f(x, y_0) - f(s, y_0)|$$

The first term is $< \frac{\epsilon}{2(b-a)}$ because $(s, y) \in U_x \times V_x$. The second term is $< \frac{\epsilon}{2(b-a)}$ because $(s, y_0) \in U_x \times V_x$ (since y_0 is the center of V_x).

So, for any $s \in U_x$ and $y \in V_x$:

$$|f(s, y) - f(s, y_0)| < \frac{\epsilon}{b - a}$$

The collection of sets $\{U_x\}_{x \in [a,b]}$ forms an open cover of $[a, b]$. Since $[a, b]$ is compact, there exists a finite subcover $U_{x_1}, U_{x_2}, \dots, U_{x_n}$. Associated with these are the corresponding neighborhoods in Y : $V_{x_1}, V_{x_2}, \dots, V_{x_n}$. We need a single δ that works for the whole interval. We take the intersection of the Y -neighborhoods:

$$V = V_{x_1} \cap V_{x_2} \cap \dots \cap V_{x_n}$$

Since this is a finite intersection of open sets containing y_0 , V is open and contains a ball of radius $\delta > 0$ around y_0 .

Let $y \in Y$ such that $d_Y(y, y_0) < \delta$ (i.e., $y \in V$). For any arbitrary $x \in [a, b]$, x must fall into one of the sets in the finite subcover, say U_{x_k} . Since $y \in V$, we know $y \in V_{x_k}$ as well. Therefore,

$$|f(x, y) - f(x, y_0)| < \frac{\epsilon}{b - a}$$

Now we can integrate to get:

$$|F(y) - F(y_0)| \leq \int_a^b |f(x, y) - f(x, y_0)| dx < \int_a^b \frac{\epsilon}{b - a} dx = \epsilon.$$

QED

Problem 5.30

Proof. (a) The equation $xe^y + ye^x = 0$ simplifies to $ye^{-y} = -xe^{-x}$. There is no way of solving this explicitly by decoupling the terms.

(b) The matrix of the total derivative is simply $[\frac{\partial f(0,0)}{\partial y}] = [e^0 + 0e^0] = [1]$ which is invertible (since its determinant is $1 \neq 0$). Hence, the Implicit Function Theorem tells us that a solution $y = y(x)$ exists. Since the function $f(x, y) = xe^y + ye^x$ is C^∞ , so is the solution $y = y(x)$.

(c) We have the equation $xe^y + ye^x = 0$, and we know y is a function of x . So we write this as $xe^{y(x)} + y(x)e^x = 0$.

We take derivatives with respect to x on both sides, to get $x \frac{dy}{dx} e^{y(x)} + e^{y(x)} + y(x)e^x + e^x \frac{dy}{dx} = 0$.

Plugging in $x = 0$, we get $e^{y(0)} + y(0) + \frac{dy}{dx}(0, 0) = 0$. Hence, $\frac{dy}{dx}(0, 0) = -e^{y(0)} - y(0)$.

We know that $y(0) = 0$ hence $\frac{dy}{dx}(0, 0) = -1$.

(d) From above, we have $x \frac{dy}{dx} e^{y(x)} + e^{y(x)} + y(x)e^x + e^x \frac{dy}{dx} = 0$. We take derivatives with respect to x on both sides.

This gives us $(xe^{y(x)} + e^x)y''(x) + xe^{y(x)}(y'(x))^2 + 2(e^{y(x)} + e^x)y'(x) + y(x)e^x = 0$.

Plugging in $x = 0$, we get $e^0 \frac{d^2 y}{dx^2}(0, 0) + 2(e^{y(0)} + e^0) \frac{dy}{dx}(0, 0) + y(0)e^0 = 0$.

Thus, $\frac{d^2 y}{dx^2}(0, 0) = -4 \frac{dy}{dx}(0, 0) = 4$.

(e) We know that $y(0) = 0, y'(0) = -1, y''(0) = 4$.

Hence it is neither a local maximum nor a local minimum.

- (f) In part (a) we found it was impossible to explicitly write down a formula $y = y(x)$ using standard functions.

Implicit Function Theorem effectively says “Don’t worry about the fact that you can’t solve for y . I will guarantee that a well-behaved function $y(x)$ exists near your point $(0,0)$, as long as you check that the matrix of partials $\frac{\partial f}{\partial y}$ is invertible”. We check exactly this in part (b) and found that the condition is met.

Without the Implicit Function Theorem, taking the derivative of $xe^y + ye^x = 0$ is just a formal, “mechanical” exercise. We’d be assuming y is a function of x without any justification, and our results might be meaningless. But with the guarantee from the Implicit Function Theorem, we know y exists and is a differentiable function. We are fully justified in using implicit differentiation.

The Implicit Function Theorem lets us do a complete local analysis (value, slope, and concavity) on a function we cannot even write down. We’ve successfully described the precise shape of a graph near the origin, all without ever knowing the $y = f(x)$ formula for it. That’s the power of the Implicit Function Theorem.

QED

Extra Problem (Counterexample for Inverse Function Theorem if f is not C^1)

Proof. $f(x) = x + x^2 \sin(\frac{1}{x})$.

Checking conditions:

1. Differentiability and $f'(0)$: Using the limit definition of the derivative, we find that $f'(0) = 1$, hence the derivative is invertible (non-zero) at the origin.
2. Discontinuity of f' : For $x \neq 0$, the derivative is $f'(x) = 1 - \cos(\frac{1}{x}) + 2x \sin(\frac{1}{x})$; since $\cos(\frac{1}{x})$ oscillates without settling as $x \rightarrow 0$, f' is not continuous at 0 (failing the C^1 condition).
3. f is not invertible: In every neighborhood of zero, there are points where $\cos(\frac{1}{x}) \approx 1$ (making the first two terms ≈ 0) and the $2x \sin(\frac{1}{x})$ term forces $f'(x)$ to take small negative values. Because $f'(x)$ takes both positive and negative values arbitrarily close to zero, f is not monotonic (and therefore not injective) in any neighborhood of the origin, so no local inverse exists.

QED